Observability (Scenarios)

Key Metrics

Metric	Type	Description
`jvm_cpu_recent_utilization_ratio`	Gauge	Recent CPU usage by JVM
`jvm_thread_count`	Gauge	Number of active JVM threads
`jvm_memory_used_bytes`	Gauge	Memory currently used by the JVM
`jvm_gc_collection_seconds_total`	Counter	Total time spent in garbage collection
`http_requests_total`	Counter	Total number of HTTP requests received
`http_server_request_duration_seconds_bucket`	Histogram	Duration of HTTP server requests
`http_client_request_duration_seconds_bucket`	Histogram	Duration of outgoing HTTP client requests
`http_errors_total`	Counter	Total number of HTTP errors (4xx, 5xx)
`process_resident_memory_bytes`	Gauge	Resident memory used by the process
`node_memory_MemAvailable_bytes`	Gauge	Available memory on the node

Scenarios

🟢 Low Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=100&faults=0' | vegeta attack -duration=60s --rate=10 |
vegeta report
```

- **Metrics to observe:**
  - `jvm_cpu_recent_utilization_ratio` : low CPU usage (max ~0.0175)
  - `jvm_thread_count` : low thread count
  - `http_server_request_duration_seconds_bucket` : stable, low latency (~3.25 ms peak)
  - `http_client_request_duration_seconds_bucket` : stable, low latency (2–6 ms)

🟡 Mid Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=100000&faults=0' | vegeta attack -duration=60s --rate=20
-timeout=120s | vegeta report
```

- **Metrics to observe:**
  - `jvm_cpu_recent_utilization_ratio` : higher CPU usage (peak ~0.118)
  - `jvm_thread_count` : increased threads, mostly waiting/runnable
  - `http_server_request_duration_seconds_bucket` : gateway and math-service show slowdowns, some 503 errors
  - `http_client_request_duration_seconds_bucket` : increased latency, some request timeouts

🔴 High Load

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=1000000&faults=0' | vegeta attack -duration=180s --
rate=60 -timeout=120s | vegeta report
```

- **Metrics to observe:**
  - `jvm_cpu_recent_utilization_ratio` : high CPU usage
  - `jvm_thread_count` : high, thread contention visible
  - `http_server_request_duration_seconds_bucket` : almost all requests fail or timeout
  - `http_client_request_duration_seconds_bucket` : extreme latency, 0% success

⚠ Mid Load with Faults (40%)

- **Vegeta command:**

```
echo 'GET http://127.0.0.1:8080/mcd?a=18&b=12&times=1000000&faults=40' | vegeta attack -duration=180s --
rate=3 -timeout=120s | vegeta report
```

- **Metrics to observe:**
  - `jvm_cpu_recent_utilization_ratio` : moderate to high CPU usage
  - `jvm_thread_count` : elevated threads
  - `http_server_request_duration_seconds_bucket` : mixed success/failures (200, 500, 503)
  - `http_client_request_duration_seconds_bucket` : most requests fail or timeout